

GAYATHRE V

Chennai, India | gayathrevaidya@gmail.com | +91 6381345775 |  GayathreVaidyanathan | 
LinkedIn |  Portfolio

Summary

Data Science final year student passionate about creating useful artificial intelligence systems through the application of LLMs and retrieval augmented generation models. I find great pleasure in turning experimentations into useful applications, always looking to build systems that are both accurate and user-friendly. I am currently exploring various problems in the area of applied artificial intelligence.

Education

Integrated M.Sc. in Data Science

Amrita Vishwa Vidyapeetham, Coimbatore

CGPA: 7.77 / 10.00

Research & Publications

Bias-Aware Toxicity Detection in Social Networks

Accepted, IECCT 2026

- Built a bias-aware NLP system to improve fairness in toxicity detection using the Jigsaw dataset.
- Combined classical ML, deep learning, and transformer models to reduce bias in identity-related text.
- Improved detection reliability across sensitive groups by reducing false positives in identity mentions.

Alzheimer's Disease Classification via Deep Learning Ensembles *Communicated, Computers in Biology and Medicine*

- Developed an ensemble-based deep learning model for MRI-based Alzheimer's detection.
- Benchmarked multiple architectures and achieved high classification performance on medical datasets.
- Used Grad-CAM to validate model focus on clinically relevant brain regions.

Projects

LegalLens — AI Contract Intelligence System

Python, LLMs, RAG, LangChain, ChromaDB, Streamlit

- Built a multi-agent RAG system to analyze contracts and extract risks, obligations, and key clauses in seconds.
- Designed an end-to-end pipeline combining semantic search and LLM reasoning, delivered through a web interface.

RepoMind — AI Codebase Intelligence Tool

Python, RAG, LangGraph, LLMs, ChromaDB

- Built a system that understands code repositories and enables natural language Q&A, bug detection, and documentation generation.
- Used semantic search and graph-based workflows to make large codebases easier to explore and analyze.

Optix — Options Strategy Visualizer & Risk Engine

Next.js, FastAPI, WebSockets, Redis, D3.js

- Built a real-time platform for analyzing multi-leg options strategies with live P&L and risk visualizations.
- Engineered fast updates using WebSockets and caching, making complex financial concepts interactive and intuitive.

Technical Skills

Languages: Python, SQL, R

ML & Data Science: Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, SciPy, SHAP, MLflow

Deep Learning & NLP: PyTorch, TensorFlow, Hugging Face, Transformers, LSTM, Vision Transformers

LLMs & RAG: LangChain, LangGraph, ChromaDB, Sentence Transformers, Prompt Engineering, Vector Search

Backend & Frontend: FastAPI, Flask, React, Next.js, Streamlit, REST APIs

Tools & Infrastructure: Docker, Redis, Git, Azure | **Data Analysis:** SQL, Excel, Data Visualisation (via Google Data Analytics Certificate)